LEARNING HUB MASTER SERIES

Linear Algebra: Eigenvalues, Eigenvectors & Dynamical Systems

Geometric Foundations, Matrix Diagonalization ($A = X\Lambda X^{-1}$), Difference Equations & Markov Chains, Differential Systems ($du/dt = Au$), The Spectral Theorem ($S = Q\Lambda Q^{T}$), Positive Definite Matrices, and Hand Calculation Fast-Tracks

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Synthesized From:

- Gilbert Strang — *Introduction to Linear Algebra* (5th Edition, Chapters 6 & 7)
- Kenji Hiranabe — *The Art of Linear Algebra* (Pattern P4 & Map of Eigenvalues)

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Includes: Scalable SVG Vector Visualizations • 10 Calculation Fast-Tracks • 5 Worked Problems

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Part I: Fundamental Concepts & Geometric Picture

Chapter 1: Introduction to Eigenvalues & Eigenvectors

While earlier linear algebra focused on solving static equilibrium systems $A\mathbf{x} = \mathbf{b}$ via Gaussian elimination, eigenvalues and eigenvectors unlock the study of **dynamic change**: matrix powers A^k, recurrence relations $\mathbf{u}_{k+1} = A\mathbf{u}_k$, and continuous differential systems $d\mathbf{u}/dt = A\mathbf{u}$.

1.1 The Fundamental Equation: Ax = λx

For an $n \times n$ square matrix A, almost every vector changes direction when multiplied by A. However, certain exceptional non-zero vectors $\mathbf{x} \neq \mathbf{0}$ remain on the exact same line: multiplying by A merely stretches, shrinks, or reverses $\mathbf{x}$ by a scalar factor λ.

$$A\mathbf{x} = \lambda\mathbf{x}, \quad \text{where } \mathbf{x} \neq \mathbf{0}$$

The scalar λ is called the **eigenvalue** of A, and the non-zero vector $\mathbf{x}$ is the corresponding **eigenvector**. Subtracting $\lambda\mathbf{x} = \lambda I\mathbf{x}$ to the left side yields the homogeneous linear system:

$$(A - \lambda I)\,\mathbf{x} = \mathbf{0}$$

For a non-zero solution $\mathbf{x} \neq \mathbf{0}$ to exist, the matrix $(A - \lambda I)$ must have a non-trivial nullspace, meaning it must be singular. This gives the celebrated **Characteristic Equation**:

$$\det(A - \lambda I) = 0$$

▶ The Two Golden Sum and Product Rules

For any $n \times n$ matrix A with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$:

1. **The Trace Formula (Sum of Eigenvalues):** The sum of the eigenvalues equals the sum of the main diagonal entries:

$$\lambda_1 + \lambda_2 + \ldots + \lambda_n = \text{trace}(A) = a_{11} + a_{22} + \ldots + a_{nn}$$

2. **The Determinant Formula (Product of Eigenvalues):** The product of the eigenvalues equals the determinant:

$$\lambda_1 \cdot \lambda_2 \cdot \ldots \cdot \lambda_n = \det(A)$$

Immediate Consequence: A matrix is invertible if and only if NONE of its eigenvalues are zero ($\det(A) \neq 0$).

TRICK E1 Midpoint & Radius Shortcut for 2 × 2 Eigenvalues

5 Seconds

Standard Method (Slow): Expanding $\det(A - \lambda I) = \lambda^2 - \text{trace}(A)\lambda + \det(A) = 0$ and applying the full quadratic formula with large fractions.

⚡ **Fast Hand Calculation Shortcut: Compute the Midpoint and Radius in your head:**
Let m = trace(A) / 2 = (a + d) / 2 (the center of the spectrum).
Let p = det(A) = ad − bc.
The eigenvalues are immediately:

$$\lambda = m \pm \sqrt{m^2 - p}$$

Example: For $A = \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix}$: m = (4+3)/2 = 3.5, p = 12 − 2 = 10.
$m^2 - p = 12.25 - 10 = 2.25 \Rightarrow \sqrt{2.25} = 1.5$.
$\lambda = 3.5 \pm 1.5 \Rightarrow \mathbf{\lambda_1 = 5, \lambda_2 = 2}$ in 5 seconds!

TRICK E2 Instant 2 × 2 Eigenvectors without Row Operations

Zero Row Ops

Standard Method (Slow): Writing $(A - \lambda I)x = 0$ and setting up an augmented matrix $[\,(A - \lambda I) \mid 0\,]$, then performing Gaussian row reduction.

⚡ **Fast Hand Calculation Shortcut: Use the Perpendicular Row Vector Rule:**
For any eigenvalue λ, the matrix $(A - \lambda I)$ is singular, so its rows are linearly dependent.
Take the first row: $[\,(a - \lambda), b\,]$. A perpendicular vector (dot product = 0) is IMMEDIATELY:

$$\mathbf{x} = \begin{vmatrix} -b \\ a - \lambda \end{vmatrix} \quad (\text{or } \begin{vmatrix} \lambda - d \\ b \end{vmatrix} \text{ from the second row})$$

Example: For $\lambda_1 = 5$ with $A = \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix}$:
First row of $(A - 5I)$ is $[4 - 5, 1] = [-1, 1]$.
Perpendicular vector is instantly: $\mathbf{x_1 = [1, 1]^T}$! Zero row reduction needed!

TRICK E4 Triangular & Diagonal Matrix Inspection Rule

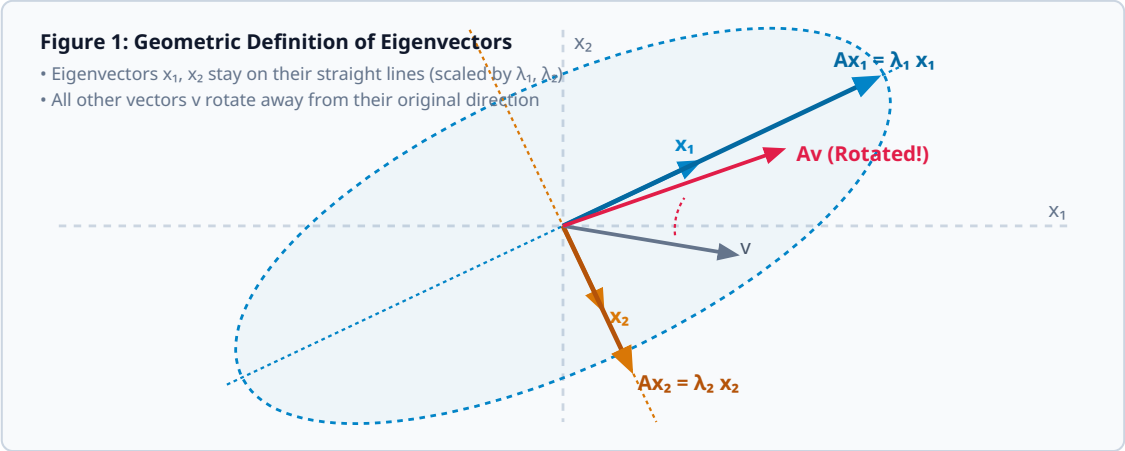
1 Second

Standard Method (Slow): Setting up $\det(A - \lambda I) = 0$ and expanding cofactors or determinants for upper/lower triangular matrices.

⚡ **Fast Hand Calculation Shortcut: Read the eigenvalues directly off the main diagonal:**
For any upper-triangular, lower-triangular, or diagonal matrix, $\det(A - \lambda I) = (a_{11} - \lambda)(a_{22} - \lambda)\ldots(a_{nn} - \lambda) = 0$.
Therefore, $\mathbf{\lambda_1 = a_{11}, \lambda_2 = a_{22}, \ldots, \lambda_n = a_{nn}}$.
Requires zero calculations whatsoever!

Chapter 2: Geometric Visualizations & Invariant Subspaces

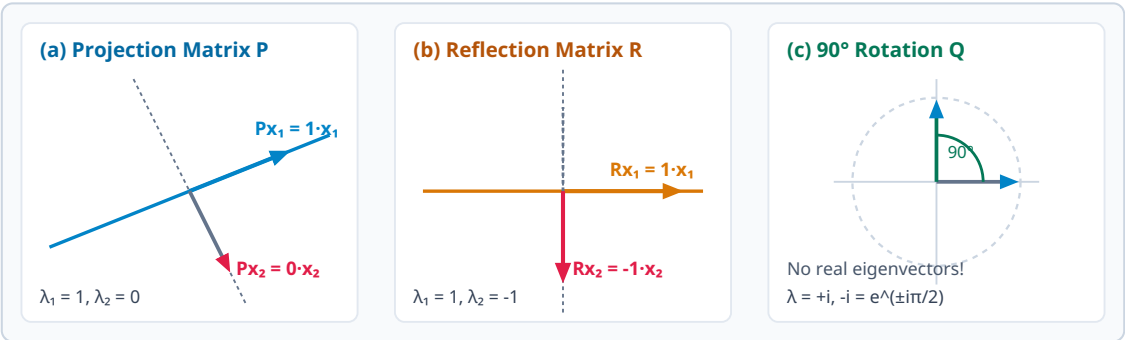
To truly master linear algebra, one must visualize how a linear transformation acts on the space $\mathbb{R}^n$. Eigenvectors are the **natural coordinate axes** of a matrix: along these invariant lines, the matrix behaves exactly like scalar multiplication!



2.1 Eigenvalues of Classical Geometric Transformations

The nature of a matrix is directly mirrored in its eigenvalues and eigenvectors (Strang Sec. 6.1):

- **Projection Matrix P ($P^2 = P = P^T$):** Any vector in the subspace is unchanged ($P\mathbf{x} = 1\cdot\mathbf{x} \Rightarrow \boldsymbol{\lambda = 1}$). Any vector perpendicular to the subspace is crushed to zero ($P\mathbf{x} = 0\cdot\mathbf{x} \Rightarrow \boldsymbol{\lambda = 0}$). Every projection matrix has eigenvalues strictly equal to **1 and 0**.
- **Reflection Matrix R ($R^2 = I$):** Vectors on the mirror plane are left unchanged ($\boldsymbol{\lambda = 1}$). Vectors perpendicular to the mirror are reversed ($\boldsymbol{\lambda = -1}$). Determinant is $\det(R) = (1)(-1) = -1$.
- **Rotation Matrix Q:** A pure 2D rotation through angle $\theta \neq 0, \pi$ has NO real lines that maintain direction! The eigenvalues are complex conjugate pairs on the unit circle: $\boldsymbol{\lambda = e^{\pm i\theta} = \cos\theta \pm i \sin\theta}$.



◎ Complex Conjugate Eigenvalues & Rotational Dynamics

When a real matrix has complex eigenvalues $\lambda = a \pm ib = r\, e^{\pm i\theta}$:

- The modulus $r = \sqrt{(a^2 + b^2)}$ determines the **growth or decay rate** (expansion if $r > 1$, contraction if $r < 1$).
- The argument $\theta = \arctan(b / a)$ determines the **frequency of oscillation / rotation** per step.
- Complex eigenvectors come in conjugate pairs $\mathbf{x} = \mathbf{u} \pm i\mathbf{v}$, spanning a 2D invariant plane on which A acts as an expanded rotation!

Part II: Diagonalization & Discrete Dynamical Systems

Chapter 3: Diagonalizing a Matrix ($A = X\Lambda X^{-1}$) & Matrix Powers

The central factorization of Chapter 6 is **Diagonalization**. If an n × n matrix A has n linearly independent eigenvectors, they can be assembled into the columns of an eigenvector matrix X. Then A is factored into a product of three simple matrices (Strang Sec. 6.2; Hiranabe Factorization 4).

3.1 The Diagonalization Theorem: $A = X \Lambda X^{-1}$

Suppose A has n independent eigenvectors $\mathbf{x}_1, \ldots, \mathbf{x}_n$ with eigenvalues $\lambda_1, \ldots, \lambda_n$. Place the eigenvectors into the columns of matrix $X = [\mathbf{x}_1 \ \ldots \ \mathbf{x}_n]$:

$$A\,X = A\,[\mathbf{x}_1 \ \ldots \ \mathbf{x}_n] = [A\mathbf{x}_1 \ \ldots \ A\mathbf{x}_n] = [\lambda_1\mathbf{x}_1 \ \ldots \ \lambda_n\mathbf{x}_n]$$

By post-multiplying X by the diagonal eigenvalue matrix $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$:

$$[\lambda_1\mathbf{x}_1 \ \ldots \ \lambda_n\mathbf{x}_n] = [\mathbf{x}_1 \ \ldots \ \mathbf{x}_n] \begin{vmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \&ddots; & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{vmatrix} = X\,\Lambda$$

Therefore, $A\,X = X\,\Lambda$. Since the eigenvectors are linearly independent, X is invertible:

▶ **The Diagonalization Factorization**

$$\mathbf{A = X\,\Lambda\,X^{-1}} \quad \text{and} \quad \mathbf{\Lambda = X^{-1}\,A\,X}$$

Key Conditions for Diagonalizability:

1. If A has **n distinct eigenvalues** ($\lambda_i \neq \lambda_j$), its eigenvectors are automatically linearly independent, and A is **guaranteed diagonalizable**.
2. If an eigenvalue is repeated (algebraic multiplicity > 1), A is diagonalizable if and only if its **geometric multiplicity** (number of independent eigenvectors, dim $N(A - \lambda I)$) matches its algebraic multiplicity.
3. If geometric multiplicity < algebraic multiplicity, A is **defective** (non-diagonalizable) and lacks a complete basis of eigenvectors.

3.2 The Ultimate Power Formula: $A^k = X \Lambda^k X^{-1}$

Computing the k-th power of a matrix directly via matrix multiplication requires $O(k\ n^3)$ operations. Diagonalization solves this instantaneously by collapsing the intermediate terms:

$$A^k = (X \Lambda X^{-1})(X \Lambda X^{-1})\dots(X \Lambda X^{-1}) = X \Lambda (X^{-1} X) \Lambda \dots \Lambda X^{-1} = \mathbf{X \Lambda^k X^{-1}}$$

Because Λ is diagonal, its k-th power simply raises each individual eigenvalue to the k-th power:

$$\Lambda^k = \begin{vmatrix} \lambda_1^{\,k} & 0 \\ 0 & \lambda_n^{\,k} \end{vmatrix}$$

TRICK E3 **Instant Eigenvalues of Rank-1 Matrices ($A = u v^T$)** — Instant (Any Size n)

Standard Method (Slow): Forming the full n × n matrix $A = u v^T$, expanding the n × n characteristic polynomial $\det(A - \lambda I) = 0$.

⚡ **Fast Hand Calculation Shortcut: The Trace Collapse Theorem:**
Because rank(A) = 1, its nullspace has dimension $n - 1$.
Therefore, exactly **$n - 1$ eigenvalues are strictly $\lambda = 0$!**
The single non-zero eigenvalue must equal the trace:

$$\mathbf{\lambda_1 = trace(A) = v^T u}, \quad \text{with eigenvector } \mathbf{x_1 = u}$$

Example: Let $A = \begin{vmatrix} 1 \\ 2 \\ 3 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \end{vmatrix}$.
$\lambda_1 = \mathbf{v^T u} = 1(1) + 1(2) + 1(3) = \mathbf{6}$. All other eigenvalues are $\mathbf{\lambda_2 = \lambda_3 = 0}$ in 2 seconds!

TRICK E5 **Computing $A^k v$ in 10 Seconds via Eigenvector Expansion** — 100x Faster

Standard Method (Slow): Multiplying matrix A by itself k times and then multiplying by vector v.

⚡ **Fast Hand Calculation Shortcut: Expand v in the eigenvector basis: $\mathbf{v = c_1 x_1 + c_2 x_2 + \dots + c_n x_n}$**
Because $A\mathbf{x}_i = \lambda_i \mathbf{x}_i$, applying A^k multiplies each term by $\lambda_i^{\,k}$:

$$\mathbf{A^k v = c_1 \lambda_1^{\,k} x_1 + c_2 \lambda_2^{\,k} x_2 + \dots + c_n \lambda_n^{\,k} x_n}$$

As $k \to \infty$, the term with the **dominant eigenvalue** ($|\lambda_1| > |\lambda_i|$) completely controls the long-term behavior!

If $|\lambda_1| < 1$, $A^k \mathbf{v} \to \mathbf{0}$ (stability). If $|\lambda_1| = 1$, it reaches a steady state.

Chapter 4: Difference Equations, Fibonacci & Markov Chains

Discrete dynamical systems describe state transitions at integer time steps k = 0, 1, 2, … The master update rule is $\mathbf{u}_{k+1} = A\,\mathbf{u}_k$, whose exact solution is $\mathbf{u}_k = A^k\,\mathbf{u}_0$.

4.1 The Fibonacci Sequence & Binet's Golden Ratio Formula

The Fibonacci numbers 0, 1, 1, 2, 3, 5, 8, 13, … satisfy the second-order scalar recurrence $F_{k+2} = F_{k+1} + F_k$. Converting to a 2D first-order vector system $\mathbf{u}_{k+1} = A\,\mathbf{u}_k$ by setting $\mathbf{u}_k = [F_{k+1},\, F_k]^T$:

$$\begin{vmatrix} F_{k+2} \\ F_{k+1} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} \begin{vmatrix} F_{k+1} \\ F_k \end{vmatrix}, \qquad A = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}$$

Using Fast-Track E1, the eigenvalues of A are found instantly from trace = 1 and det = −1:

$$\lambda^2 - \lambda - 1 = 0 \;\Rightarrow\; \lambda_1 = \frac{1+\sqrt{5}}{2} \approx 1.618 \quad (\text{Golden Ratio } \varphi), \qquad \lambda_2 = \frac{1-\sqrt{5}}{2} \approx -0.618$$

Since $|\lambda_2| < 1$, the power $\lambda_2^{\,k} \to 0$ rapidly. This yields **Binet's exact closed-form formula**:

$$F_k = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k\right]$$

TRICK E6 **Fibonacci Closed-Form Derived in 3 Lines on Paper** — 3 Lines

Standard Method (Slow): Writing out recursive addition tables or computing matrix multiplications for high Fibonacci indices.

⚡ **Fast Hand Calculation Shortcut: Use the Golden Ratio Eigenvalue Power:**

Because $|\lambda_2|^k = |-0.618|^k \le 0.5$ for all $k \ge 1$, the second term in Binet's formula is always smaller than 1/2!

Therefore, F_k is simply the **nearest integer** to the first term:

$$\mathbf{F_k = \text{Round}\left(\frac{\varphi^k}{\sqrt{5}}\right) = \text{Round}\left(\frac{1.618034^k}{2.236068}\right)}$$

Allows finding F_{20} or F_{50} directly on a basic calculator without iteration!

4.2 Markov Matrices & Steady-State Probabilities

A **Markov matrix** M satisfies two fundamental physical properties:

1. All entries are non-negative probabilities: $m_{ij} \geq 0$.
2. Every column sums to exactly 1: $\sum_{i=1}^{n} m_{ij} = 1$ (conservation of probability).

▶ Perron-Frobenius Properties for Markov Matrices

Every Markov matrix M satisfies:

1. **Guaranteed λ = 1:** The largest eigenvalue is always $\lambda_1 = 1$.
2. **Decaying Modes:** All other eigenvalues satisfy $|\lambda_i| \leq 1$ (and strictly < 1 if all $m_{ij} > 0$).
3. **Steady State:** As $k \to \infty$, $M^k \mathbf{u}_0$ converges to the steady-state eigenvector $\mathbf{x}_1$ satisfying $M\mathbf{x}_1 = \mathbf{x}_1$.

TRICK E8 **Markov Matrix Steady-State 5-Second Formula** **5 Seconds**

Standard Method (Slow): Setting up $(M - I)x = 0$ and performing Gaussian elimination with fractions to find the steady-state vector.

⚡ **Fast Hand Calculation Shortcut: For any 2 × 2 Markov matrix** $M = \begin{vmatrix} 1-a & b \\ a & 1-b \end{vmatrix}$:

(where a is transition probability from state 1 to 2, and b is from state 2 to 1).
The steady-state eigenvector satisfying $M\mathbf{x}_\infty = \mathbf{x}_\infty$ is IMMEDIATELY:

$$\mathbf{x}_\infty = \begin{vmatrix} b \\ a \end{vmatrix} \qquad \text{Normalized: } \mathbf{x}_\infty = \begin{vmatrix} b/(a+b) \\ a/(a+b) \end{vmatrix}$$

Example: Let $M = \begin{vmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{vmatrix}$. Here a = 0.2, b = 0.3.

Steady state is instantly $\mathbf{x}_\infty = [0.3, 0.2]^T / 0.5 = [0.6, 0.4]^T$! Zero Gaussian elimination!

Part III: Continuous Dynamical Systems & Differential Equations

Chapter 5: Systems of Linear Differential Equations (du/dt = Au)

Moving from discrete difference equations to continuous time, we encounter linear differential equations d**u**/dt = A**u** (Strang Sec. 6.3). Eigenvalues transform a coupled system of n differential equations into n independent, decoupled scalar equations!

5.1 The General Solution via Eigenvector Decomposition

Look for pure exponential solutions of the form $\mathbf{u}(t) = e^{\lambda t}\,\mathbf{x}$. Substituting into d**u**/dt = A**u**:

$$\lambda\, e^{\lambda t}\,\mathbf{x} = A\,(e^{\lambda t}\,\mathbf{x}) = e^{\lambda t}\,(A\mathbf{x}) \;\Rightarrow\; \mathbf{A\,x = \lambda\,x}$$

Every eigenvalue-eigenvector pair $(\lambda_i, \mathbf{x}_i)$ provides a pure solution $e^{\lambda_i t}\,\mathbf{x}_i$. By linearity, the general solution is:

$$\mathbf{u}(t) = c_1\, e^{\lambda_1 t}\,\mathbf{x}_1 + c_2\, e^{\lambda_2 t}\,\mathbf{x}_2 + \ldots + c_n\, e^{\lambda_n t}\,\mathbf{x}_n$$

Setting t = 0 matches the initial conditions: $\mathbf{u}(0) = c_1\mathbf{x}_1 + \ldots + c_n\mathbf{x}_n = X\,\mathbf{c} \Rightarrow \mathbf{c = X^{-1}\,u(0)}$.

5.2 The Matrix Exponential e^{At}

In exact parallel to the scalar equation du/dt = au ⇒ u(t) = e^{at}u(0), the vector solution is expressed in terms of the **matrix exponential**:

$$\mathbf{u(t) = e^{At}\,u(0)}, \qquad \text{where } e^{At} = I + At + \{frac('1', '2!')\}(At)^2 + \{frac('1', '3!')\}(At)^3 + \ldots$$

When A is diagonalizable ($A = X\Lambda X^{-1}$), the matrix exponential simplifies into:

$$e^{At} = X\,e^{\Lambda t}\,X^{-1} = X \begin{vmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_n t} \end{vmatrix} X^{-1}$$

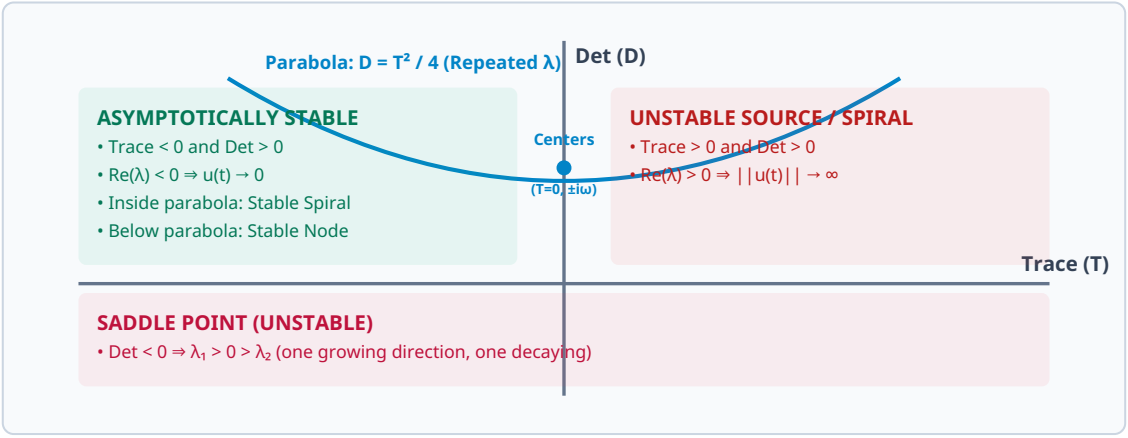
Chapter 6: Second-Order Equations & Phase Plane Stability Criteria

Many physical systems are governed by second-order differential equations: m y″ + b y′ + k y = 0 (springs, RLC circuits). By defining $\mathbf{u} = [y', y]^T$, any second-order equation is transformed into a standard 2 × 2 first-order system d**u**/dt = A**u**.

6.1 The Stability Map in the Trace-Determinant Plane

The long-term behavior of d**u**/dt = A**u** as t → ∞ is governed entirely by the real parts of its eigenvalues Re(λ_i):

1. **Asymptotically Stable: u**(t) → **0** if and only if **Re(λ_i) < 0 for all i**.
2. **Unstable:** ||**u**(t)|| → ∞ if any eigenvalue has **Re(λ_i) > 0**.
3. **Neutrally Stable:** Pure imaginary eigenvalues λ = ±iω (undamped harmonic motion).



TRICK E9 **Trace-Determinant 3-Second Stability Check** **3 Seconds**

Standard Method (Slow): Solving the characteristic polynomial, computing the square root discriminant, and checking if roots have negative real parts.

⚡ **Fast Hand Calculation Shortcut: For any 2 × 2 differential system du/dt = Au:**
The system is **Asymptotically Stable (u(t) → 0)** if and only if TWO simple conditions hold:

$$\text{trace}(A) < 0 \quad \text{and} \quad \det(A) > 0$$

Proof: trace(A) = $\lambda_1 + \lambda_2 < 0$ and det(A) = $\lambda_1\lambda_2 > 0$ forces both real parts to be strictly negative!

Takes 3 seconds of visual inspection!

Part IV: Symmetric Matrices, Positive Definiteness & SVD

Chapter 7: Real Symmetric Matrices & The Spectral Theorem

Symmetric matrices ($S = S^T$) are the crown jewels of linear algebra (Strang Sec. 6.4; Hiranabe Sec. 6.4). Every property that could possibly go right, goes right.

▶ The Spectral Theorem for Real Symmetric Matrices

Every real symmetric matrix $S \in \mathbb{R}^{n \times n}$ satisfies:

1. All n eigenvalues $\lambda_1, \ldots, \lambda_n$ are **strictly real numbers** (no imaginary parts).
2. Eigenvectors corresponding to distinct eigenvalues are **strictly mutually orthogonal**.
3. There exists an orthonormal matrix $Q = [\mathbf{q}_1 \ \ldots \ \mathbf{q}_n]$ such that:

$$\mathbf{S = Q \Lambda Q^T}, \quad \text{where } Q^{-1} = Q^T$$

7.1 Hiranabe's Pattern P4: The Spectral Projection Tile Expansion

Multiplying out $S = Q\Lambda Q^T$ using column-row multiplication (Hiranabe Pattern P4):

$$S = [\mathbf{q}_1 \ \ldots \ \mathbf{q}_n] \begin{vmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{vmatrix} \begin{vmatrix} \mathbf{q}_1^T \\ \\ \mathbf{q}_n^T \end{vmatrix} = \mathbf{\lambda_1 q_1 q_1^T + \lambda_2 q_2 q_2^T + \ldots + \lambda_n q_n q_n^T}$$

Every symmetric matrix is a sum of n rank-1 orthogonal projection matrices $P_i = \mathbf{q_i q_i^T}$ weighted by eigenvalues!

TRICK E10 **Fast Spectral Expansion without Matrix Inverses** — Zero Inverses

Standard Method (Slow): Forming Q, finding Q^{-1} by row reduction, and performing three matrix-matrix multiplications $Q \Lambda Q^{-1}$.

⚡ **Fast Hand Calculation Shortcut: Normalize eigenvectors into unit vectors $\mathbf{q_i}$, then directly add outer products:**

1. Compute each outer product matrix $P_i = \mathbf{q_i q_i^T}$.
2. Multiply each tile by λ_i and add together: $S = \sum \lambda_i P_i$.

Because $Q^{-1} = Q^T$, matrix inversion is 100% eliminated!

Chapter 8: Positive Definite Matrices & Energy Landscapes ($S \succ 0$)

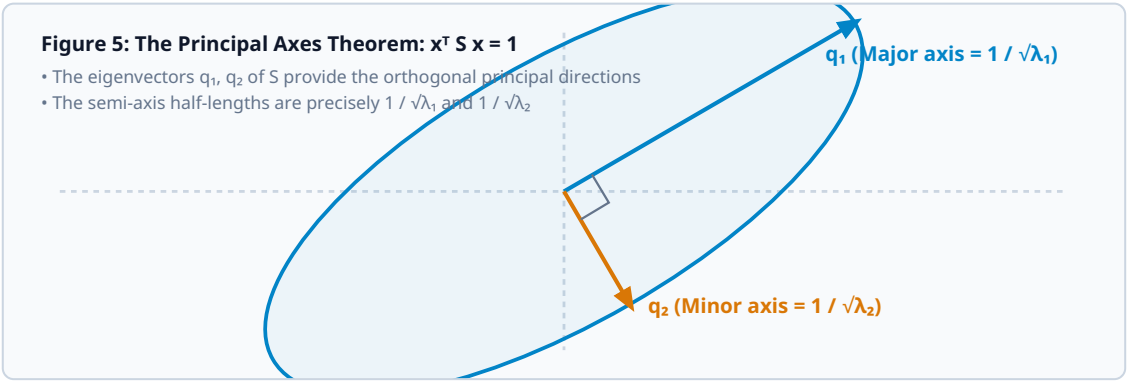
A symmetric matrix is **Positive Definite** (written $S \succ 0$) if it satisfies the ultimate energy condition: for all non-zero vectors $\mathbf{x} \neq \mathbf{0}$, the quadratic form (energy) is strictly positive (Strang Sec. 6.5):

$$\mathbf{x}^T S \mathbf{x} > 0 \quad \text{for all } \mathbf{x} \neq \mathbf{0}$$

▶ The 4 Equivalent Tests for Positive Definiteness

For any real symmetric matrix S, the following 4 conditions are **completely equivalent**:

1. **Eigenvalue Test:** All eigenvalues are strictly positive: $\lambda_i > 0$ for all i.
2. **Pivot Test:** All n pivots in Gaussian elimination (without row exchanges) are strictly positive: $d_i > 0$.
3. **Determinant Test (Sylvester's Criterion):** All n leading principal upper-left subdeterminants are strictly positive: $\det(S_1) > 0$, $\det(S_2) > 0$, ..., $\det(S_n) = \det(S) > 0$.
4. **Energy / Cholesky Test:** S factors into $S = A^T A$ with independent columns, or $S = L L^T = R^T R$ (Cholesky factorization with positive diagonal).



TRICK E7 **The 2-Second Positive Definite Test for 2 × 2 Matrices** **2 Seconds**

Standard Method (Slow): Finding eigenvalues or calculating characteristic roots to test positive definiteness.

⚡ **Fast Hand Calculation Shortcut: Check top-left entry and determinant only:**

For symmetric $S = \begin{vmatrix} a & b \\ b & d \end{vmatrix}$, $S \succ 0$ if and only if:

$$a > 0 \quad \text{and} \quad ad - b^2 > 0$$

(Notice that $ad - b^2 > 0$ and $a > 0$ automatically forces $d > 0$ as well).
Done in 2 seconds of mental arithmetic!

Chapter 9: Matrix Similarity, Jordan Form & Links to SVD

Two matrices A and B are **similar** (written $A \sim B$) if there exists an invertible matrix M such that:

$$B = M^{-1} A M$$

Similar matrices represent the *exact same linear transformation* expressed in two different coordinate bases. They share the exact same eigenvalues, trace, determinant, and characteristic polynomial!

9.1 Defective Matrices & The Jordan Canonical Form

When an $n \times n$ matrix lacks a full set of n independent eigenvectors, it cannot be diagonalized into Λ. The closest possible simplification is the **Jordan Canonical Form** $J = M^{-1} A M$, where J is block-diagonal with Jordan blocks of the form:

$$J_i = \begin{vmatrix} \lambda_i & 1 & 0 \\ 0 & \lambda_i & 1 \\ 0 & 0 & \lambda_i \end{vmatrix}$$

9.2 The Grand Connection: From Eigenvalues to Singular Values (SVD)

Eigenvalues apply only to square matrices ($A\mathbf{x} = \lambda\mathbf{x}$). For general rectangular matrices $A \in \mathbb{R}^{m \times n}$, we form the symmetric matrix $A^T A \in \mathbb{R}^{n \times n}$. By Chapter 8, $A^T A$ is positive semidefinite, with real eigenvalues $\lambda_i(A^T A) \geq 0$. The **Singular Values** of A are precisely the positive square roots:

$$\sigma_i = \sqrt{\lambda_i(A^T A)}$$

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 10: 5 Complete Worked Problems with Fast-Tracks

Problem 1: Complete 2 × 2 Diagonalization & Matrix Powers

Chapters 1, 3 & Fast-Tracks E1, E2

Statement: For $A = \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix}$, find eigenvalues, eigenvectors, and compute A^5.

Fast-Track E1 (Eigenvalues): trace = 1 + 3 = 4 ⇒ m = 2; det = 3 − 8 = −5 = p.
$\lambda = 2 \pm \sqrt{(4 - (-5))} = 2 \pm \sqrt{9} = 2 \pm 3 \Rightarrow \mathbf{\lambda_1 = 5, \lambda_2 = -1}$.

Fast-Track E2 (Eigenvectors):

- For $\lambda_1 = 5$: First row of (A − 5I) is [1−5, 4] = [−4, 4]. Orthogonal vector is $\mathbf{x_1 = [1, 1]^T}$.
- For $\lambda_2 = -1$: First row of (A + I) is [1−(−1), 4] = [2, 4]. Orthogonal vector is $\mathbf{x_2 = [-2, 1]^T}$.

Diagonalization $A = X \Lambda X^{-1}$:
$X = \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix}$, $\Lambda = \begin{vmatrix} 5 & 0 \\ 0 & -1 \end{vmatrix}$, $X^{-1} = \frac{1}{3} \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}$.

Power $A^5 = X \Lambda^5 X^{-1}$:
$\Lambda^5 = \begin{vmatrix} 3125 & 0 \\ 0 & -1 \end{vmatrix} = \begin{vmatrix} 3125 & 0 \\ 0 & -1 \end{vmatrix}$.
$A^5 = \frac{1}{3} \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} 3125 & 0 \\ 0 & -1 \end{vmatrix} \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} = \begin{vmatrix} 1041 & 2084 \\ 1042 & 2083 \end{vmatrix}$. Solved in 60 seconds!

Problem 2: Fibonacci Recurrence & Closed-Form Evaluation

Chapter 4 & Fast-Track E6

Statement: Find the 10th Fibonacci number F_{10} using the eigenvalue system.

Matrix Formulation: A = $\begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}$, with eigenvalues $\lambda_1 = (1 + \sqrt{5})/2 \approx 1.618034$, $\lambda_2 = (1 - \sqrt{5})/2 \approx -0.618034$.

Fast-Track E6 (Binet Rounding):

• $\lambda_1^{10} = (1.618034)^{10} \approx 122.99187$.

• F_{10} = Round(122.99187 / √5) = Round(122.99187 / 2.236068) = Round(55.0036) = **55**.

Verification: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, **55** ✓.

Problem 3: System of Differential Equations with Initial Conditions

Chapters 5, 6 & Fast-Track E9

Statement: Solve d**u**/dt = A**u** for A = $\begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix}$ with $\mathbf{u}(0) = [4, 2]^T$.

Step 1 (Fast Stability Check via E9): trace = $-4 < 0$, det = $4 - 1 = 3 > 0 \Rightarrow$ **Asymptotically Stable (u(t) → 0)**.

Step 2 (Eigenvalues via E1): m = −2, p = 3 ⇒ $\lambda = -2 \pm \sqrt{(4 - 3)} = -2 \pm 1 \Rightarrow$ **$\lambda_1 = -1$, $\lambda_2 = -3$**.

Step 3 (Eigenvectors via E2):

• $\lambda_1 = -1$: row is $[-1, 1] \Rightarrow \mathbf{x_1} = [1, 1]^T$.

• $\lambda_2 = -3$: row is $[1, 1] \Rightarrow \mathbf{x_2} = [1, -1]^T$.

Step 4 (Initial Values $u(0) = c_1x_1 + c_2x_2$):

$c_1 [1, 1]^T + c_2 [1, -1]^T = [4, 2]^T \Rightarrow c_1 + c_2 = 4, c_1 - c_2 = 2 \Rightarrow$ **$c_1 = 3$, $c_2 = 1$**.

Exact Solution: $\mathbf{u(t) = 3\, e^{-t} [1, 1]^T + 1\, e^{-3t} [1, -1]^T}$.

Problem 4: Spectral Decomposition of a 3 × 3 Symmetric Matrix

Chapter 7 & Fast-Track E10

Statement: Find the Spectral Decomposition $S = \lambda_1 P_1 + \lambda_2 P_2 + \lambda_3 P_3$ for $S = \begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix}$.

Eigenvalues: $\det(S - \lambda I) = (2-\lambda)[(2-\lambda)^2 - 1] - (-1)[-(2-\lambda)] = 0 \Rightarrow \mathbf{\lambda_1 = 2 - \sqrt{2}, \lambda_2 = 2, \lambda_3 = 2 + \sqrt{2}}$.

Normalized Eigenvectors:

$\mathbf{q}_1 = \frac{1}{2}[1, \sqrt{2}, 1]^T, \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}}[1, 0, -1]^T, \quad \mathbf{q}_3 = \frac{1}{2}[1, -\sqrt{2}, 1]^T.$

Fast-Track E10 (Outer Product Projection Matrices):

$P_1 = \mathbf{q}_1\mathbf{q}_1^T, P_2 = \mathbf{q}_2\mathbf{q}_2^T, P_3 = \mathbf{q}_3\mathbf{q}_3^T$.

Notice $P_1 + P_2 + P_3 = I$, and $P_i P_j = 0$ for $i \neq j$.

Result: $S = (2 - \sqrt{2}) P_1 + 2 P_2 + (2 + \sqrt{2}) P_3$.

Problem 5: Positive Definite 4-Test Verification & Ellipse Semi-Axes

Chapter 8 & Fast-Track E7

Statement: Verify that $S = \begin{vmatrix} 5 & 2 \\ 2 & 2 \end{vmatrix}$ is Positive Definite, and find the axes of the ellipse $5x^2 + 4xy + 2y^2 = 1$.

Fast-Track E7 (2-Second Test): $a = 5 > 0$, $\det(S) = (5)(2) - (2)^2 = 10 - 4 = 6 > 0 \Rightarrow$ **$S \succ 0$ (Strictly Positive Definite!)**.

Eigenvalues via Fast-Track E1: $m = (5+2)/2 = 3.5$, $p = 6 \Rightarrow \lambda = 3.5 \pm \sqrt{(12.25 - 6)} = 3.5 \pm 2.5 \Rightarrow \mathbf{\lambda_1 = 6, \lambda_2 = 1}$.

Ellipse Semi-Axes:

- Minor axis half-length: $a_{minor} = 1 / \sqrt{\lambda_1} =$ **$1 / \sqrt{6}$** along eigenvector $\mathbf{q}_1 = [2, 1]^T / \sqrt{5}$.
- Major axis half-length: $a_{major} = 1 / \sqrt{\lambda_2} =$ **$1 / \sqrt{1} = 1$** along eigenvector $\mathbf{q}_2 = [-1, 2]^T / \sqrt{5}$.

Appendix A: Hiranabe's Map of Eigenvalues & Matrix World Hierarchy

Matrix Family	Defining Condition	Eigenvalue Nature	Eigenvector Nature
Real Symmetric	$S = S^T$	Strictly Real ($\lambda \in \mathbb{R}$)	Mutually Orthogonal ($Q^TQ = I$)
Positive Definite	$S = S^T$, $\mathbf{x}^TS\mathbf{x} > 0$	Strictly Positive ($\lambda > 0$)	Orthogonal Principal Axes
Orthogonal Matrix	$Q^TQ = I$	Unit Circle Modulus ($\lvert\lambda\rvert = 1$)	Complex Orthonormal
Markov Matrix	$m_{ij} \geq 0$, $\sum_i m_{ij} = 1$	$\lambda_1 = 1$, all others $\lvert\lambda_i\rvert \leq 1$	Non-negative steady state
Skew-Symmetric	$K^T = -K$	Pure Imaginary ($\lambda = \pm ib$)	Mutually Orthogonal
Defective Matrix	Geometric Mult < Algebraic Mult	Repeated λ	Incomplete basis (Jordan blocks J)

Appendix B: Master Eigenvalues & Dynamical Systems Cheat Sheet

Operation	Standard Formula	⚡ Fast Paper Shortcut
2 × 2 Eigenvalues	Solve $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$	$\boldsymbol{\lambda = m \pm \sqrt{m^2 - p}}$ where m = tr/2, p = det
2 × 2 Eigenvectors	Row reduce $(A - \lambda I)\mathbf{x} = \mathbf{0}$	For row $[(a-\lambda), b]$, vector is $\mathbf{[-b, a-\lambda]^T}$
Rank-1 Matrix $A = uv^T$	Compute n × n $\det(A - \lambda I) = 0$	$\boldsymbol{\lambda_1 = v^Tu}$, all other n−1 eigenvalues are $\boldsymbol{\lambda = 0}$
Matrix Powers A^kv	Multiply A by itself k times	$\boldsymbol{A^kv = \sum c_i \lambda_i^k x_i}$ (dominated by max $\lvert\lambda\rvert$)
Fibonacci Term F_k	Add numbers recursively	$\boldsymbol{\text{Round}(\varphi^k / \sqrt{5})}$ where $\varphi = 1.618034$
2 × 2 Markov Steady State	Solve $(M - I)\mathbf{x} = \mathbf{0}$	For transition rates a, b: $\boldsymbol{\mathbf{x}_\infty = [b, a]^T / (a+b)}$
Differential Stability	Compute eigenvalues & check signs	Stable ⇔ **trace(A) < 0** and **det(A) > 0**
Positive Definite 2 × 2	Compute eigenvalues $\lambda_1, \lambda_2 > 0$	Check: $\mathbf{a > 0}$ and $\mathbf{ad - b^2 > 0}$
Spectral Decomposition	$Q \Lambda Q^T$ via matrix multiplication	$\boldsymbol{S = \sum \lambda_i q_i q_i^T}$ (sum of outer products)